

Letter to the editors

Response to Reviewers — Submission 40

AutoGraphForge: Towards Automated Graph Theory Discovery

Dear Editors,

We thank both reviewers for their careful reading. Below is a complete account of the changes.

1. Changes responding to Reviewer 1

A1 — Grammar (comment 3)

*“counterexample search algorithms **is** built upon”*
→ *“counterexample search algorithms **are** built upon”*

A2 — Grammar (comment 4)

*“the proving stage is **implemented in and** sanity-checked but not yet properly evaluated”*
→ *“the proving stage is **implemented and** sanity-checked but not yet properly evaluated”*

A3 — Sophie necessary conditions, readability (comment 5)

The long clause chain was replaced by short sentences leading with the mechanism:

“Sophie necessary conditions are handled by contraposition. Such a condition has the form $A \Rightarrow \neg C$, where A is a numeric predicate and C a graph class. It is flagged known whenever the contrapositive $C \Rightarrow \neg A$ appears in the table. For instance $(\chi > 2) \Rightarrow \neg \text{bipartite}$ is recognised as the contrapositive of the known bound $\text{bipartite} \Rightarrow \chi \leq 2$.”

A4 — Qualitative summary of the inspected conjectures (comment 2) — ADDED

A 35-line commented-out draft (a hand-classification of the top 50: 22/12/6/5/5) was **removed** and replaced with a live paragraph and a new table:

Category	Count
Rediscovery recognised by the novelty filter	49
Subsumed by a stronger survivor	2
Decorative hypothesis (class does no work)	1
Promising, universal	28
Promising, class-conditioned	20
Total	100

The categories are now computed by `tools/audit_sample.py` from the pipeline’s own novelty table, subsumption lattice and support statistics rather than assigned by hand, so the classification is reproducible from the released artefacts.

We report no “probably false” category and say so explicitly: every item survived refutation against the full dataset, so we have no evidence of falsity for any of them, and asserting one would be an unsupported guess.

2. Changes responding to Reviewer 2

B1 — SAT/CP-SAT for the NP-hard invariants — ADDED

Absent from the submitted version; added after the CPU-budget accounting:

“Since the precomputation dominates, and within it the NP-hard invariants dominate, replacing the ILP formulations with a SAT or CP-SAT encoding is the single most promising engineering change available to us. Independence, domination and zero-forcing all admit natural propositional encodings, and modern conflict-driven and local-search solvers are frequently far faster than ILP on instances of this size. A cheaper battery is not merely a saving: it would let us extend the refutation tiers to larger orders, and so reach invariants that are uninformative on the small graphs the present pool is concentrated on. We are grateful to a reviewer for this suggestion.”

3. Corrections

C1 — Survivor count (4 sites: abstract, Table 5 caption, §4.1, §5)

The merge step previously removed cross-partition duplicates and nothing else, so the reported figure still counted conjectures that a witness found in another partition had already refuted. The merge now applies those witnesses across partitions before counting (see C3). The arithmetic makes the difference explicit: $8,281 - 346 = 7,935$ was correct for the old merge, and $8,281 - 1,413 - 346 = 6,522$ for the corrected one.

“7,935 surviving conjectures” $\rightarrow$ “6,522 surviving conjectures”

C2, C3 and C4 follow from this change.

C2 — Survivor split (§4.1)

The stated split summed to the old total:

*“split into 3,386 class-conditioned inequalities and 4,549 Sophie **sufficient**-conditions”
 $\rightarrow$ “split into 2,677 class-conditioned inequalities and 3,845 Sophie conditions”*

$$2,677 + 3,845 = 6,522.$$

C3 — Merge description (Table 5 caption)

The whole reduction was attributed to deduplication, which accounts for only 346 of it:

*“the five sum to the 8,281 raw survivors that the merge step **deduplicates to** 7,935.”
 $\rightarrow$ “the five sum to the 8,281 raw survivors. The merge step reduces these to 6,522 **in two stages**: 1,413 **are refuted by witnesses found in another partition** — each partition grows its own hard seed and never sees the others’, so a graph that kills a candidate may sit in the wrong shard — **and a further 346 are cross-partition duplicates.**”*

C4 — Ranks in §4.2

The touch heuristic previously ignored the class hypothesis, so different hypotheses over the same body received identical touch counts:

*“the bipartite class (rank 99 of 7,935 by touch, touch 2,053) and the cubic/regular classes (cubic: rank 72, touch 2,181; regular: rank 100, touch 2,053)”
 $\rightarrow$ “the bipartite class (rank 91 by touch, touch 922) and the regular class (rank 177, touch 589). Ranks are among the 2,677 survivors that carry a recomputed, hypothesis-aware touch count; the cubic specialisation (rank 453, touch 233) is now flagged as subsumed by the regular form, since $\text{cubic} \subset \text{regular}$ ”*

C5 — Boolean count (§3.4)

The paragraph said 14 booleans and then 16, without explanation:

*“while the 16 booleans act as Sophie hypotheses and as class conditions”
 $\rightarrow$ “while the 14 booleans — **together with two derived order thresholds** ($n \geq 3$ and*

n ≥ 4) that the battery adds precisely so that bounds failing only on the smallest graphs become true class-restricted statements — give the 16 predicates available to Sophie as hypotheses and class conditions”

C6 — Round-one expansion (§4.1)

“a ~20–28% expansion of T” → “a ~17–28% expansion of T”

Per partition against $T = 2,860$: 27.7, 17.4, 17.3, 19.7, 20.3 %. The smallest partition adds 494 witnesses, i.e. 17.3 %, so the lower end was overstated.

C7 — Rediscovery counts (§3.3 and §4.1, two sites)

The manuscript reported 33 rediscoveries alongside a top-100 audit classifying 49 as recognised by the novelty filter (Table 2). 49 cannot sit inside 33. The two were measured against different versions of the novelty table, and both sites now say so. At §4.1, added:

“This count is the one recorded during the run, against the novelty table as it then stood. The table has since been extended to its present 559 entries, and re-classifying the survivors against the extended table flags 146 of the 2,677 ranked survivors as known — among them the 49 of Table 2. The two figures are therefore not comparable: 33 measures what the run itself caught, 146 what the current filter catches.”

At §3.3, the forward reference was corrected to name the run-time count and the current one.

4. Other changes

D1 — Acknowledgments

Added: “I also thank anonymous reviewers for helpful suggestions and comments.”

Added: “I gratefully acknowledge financial support for this research from the Comenius University (Grant No. UK/1020/2026).”

D2 — Code availability (addresses Reviewer 1, comment 1)

Added: “The complete list of surviving conjectures is published there as `results/survivors_by_touch.txt`.”

D3 — Declaration on Generative AI

Rewritten to the activity-taxonomy form of the CEUR template, naming the tool and the activities, and disclosing that the same tool was additionally used as a coding and experimentation assistant.

5. Disclosure: revisions independent of the review

The manuscript also incorporates work completed after submission and unrelated to the reviewers’ comments. Most substantively, the novelty filter was expanded:

“The table holds 203 entries” → “the list holds 559 entries”

with an accompanying change in how redundant candidates are treated:

*“the candidate is provably a structural consequence of the tabled theorems and is **discarded**”
→ “... and is **flagged as a known rediscovery** — excluded from the set of novel survivors, but stored, so it can serve as a soundness check”*

and a strengthened soundness condition on the admissible safe lower bounds. We note these so that the change record is complete.

The required edits increased the number of pages slightly. I apologize for this and hope it does not cause any problems.

Yours sincerely,

Ján Pastorek